

ALGEBRA 2 REVIEW

Name: _____

Date: _____ Per: _____

WEEK 2

SET **A**

1. What is the solution set of the equation below?

$$|x+4| = 3x-12$$

$$x+4 = 3x-12 \quad x+4 = -3x+12$$

$$16 = 2x \quad -8 = -4x$$

$$x = 8 \quad x = 2$$

- A. $\emptyset$
B. $\{5\}$
C. $\{8\}$
D. $\{5, 8\}$

C

2. What is the value of the expression below when $m = -7$ and $n = 4$?

$$|m-n| - m^2$$

$$|-7-4| - (-7)^2$$

$$|-11| - 49$$

$$11 - 49$$

-38

3. Which statement is justified by the commutative property?

- A. $c(a+b)^2 = c(b+a)^2$
B. $c(a+b)^2 = c(a+b)(a+b)$
C. $c(a+b) = (ac+bc)$
D. $(ab)c^2 = a(bc^2)$

A

4. What is the solution to the equation below?

$$\frac{5}{2}(8x-18)+39=2(10x+3)$$

$$20x-45+39=20x+6$$

$$20x-6=20x+6$$

$$-6 \neq 6$$

- A. $x = -3$
B. $x = 0$
C. No Solution
D. All Real Numbers

C

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SET **B**

1. Simplify the expression below.

$$\frac{10(-5-3)^2}{4-2 \cdot 3^3}$$

$$\frac{10(-8)^2}{4-2 \cdot 27}$$

$$\frac{640}{-50}$$

- A. 12.8
B. -12.8
C. 0.8
D. -0.8

B

2. Beth knows the perimeter, P , and width, w , of a rectangular window in her living room. Which formula can she use to find the length, l , of the window?

$$P = 2l + 2w$$

- A. $l = 2P - 2w$ C. $l = 2P - w$
B. $l = \frac{P}{2} - 2w$ D. $l = \frac{P-2w}{2}$

D

3. Ari, Nick, and Carlos sold candy bars for a school fundraiser. Ari sold two less than five times the number of candy bars than Carlos. Nick sold fifteen more than twice the number of candy bars than Carlos. If c is the number of candy bars that Carlos sold, how many more candy bars did Ari sell than Nick?

$$A = 5c - 2$$

$$N = 2c + 15$$

$$(5c-2) - (2c+15)$$

$$5c-2-2c-15$$

- A. $3c - 17$ C. $3c + 17$
B. $3c + 13$ D. $3c - 13$

A

4. What is the solution set of the equation below?

$$-18 = 2a - 2|1-3a|$$

$$-2a-18 = -2|1-3a|$$

$$a+9 = |1-3a|$$

$$a+9 = 1-3a \quad -a-9 = 1-3a$$

$$4a = -8 \quad -10 = -2a$$

$$a = -2 \quad 5 = a$$

- A. $\emptyset$
B. $\{-2\}$
C. $\{5\}$
D. $\{-2, 5\}$

D

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